

Research paper

Trends in depressive and anxiety disorders among adolescents and young adults (aged 10–24) from 1990 to 2021: A global burden of disease study analysis

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ABSTRACT

Objective: To assess long-term trends in the burden of depression and anxiety among individuals aged 10–24 years from 1990 to 2021, and to examine disparities across socio-demographic levels using data from the Global Burden of Disease (GBD) Study 2019 and 2021.

Methods: We performed a retrospective ecological trend analysis using Joinpoint regression models to estimate the Average Annual Percent Change (AAPC) in incidence and disability-adjusted life years (DALYs). Inequalities across countries were evaluated using the slope index of inequality (SII) and the concentration index (CI). Analyses were stratified by age group, sex, geographic region, country, and Socio-demographic Index (SDI) level. **Results:** Between 1990 and 2021, the global AAPC of depression and anxiety disorders ranged from 0.80 % to 0.97 %, with a marked acceleration from 2014 to 2021, peaking in 2019. The burden was consistently higher in females, though the growth rate of male depression was steeper. The fastest increase in depression was observed in the 10–14 age group, and in anxiety among those aged 20–24. Regionally, high-income North America showed the greatest increase in depression, while Andean Latin America exhibited the highest rise in anxiety. High SDI countries experienced greater burden growth, whereas East Asia saw the slowest. Although relative inequality decreased, absolute inequality across countries widened over time.

Conclusions: The burden of depression and anxiety among adolescents and young adults has increased substantially, especially in high SDI countries. These trends vary significantly by sex, age, and region, highlighting the need for context-specific mental health interventions.

1. Introduction

It is estimated that 1.1 % of adolescents aged 10–14 years and 2.8 % of those aged 15–19 years globally experience depression, while 3.6 % of adolescents aged 10–14 years and 4.6 % of those aged 15–19 years suffer from anxiety disorders (World Health Organization, 2021). The incidence of depression and anxiety significantly increases during adolescence, particularly in early adolescence, with approximately 3.1 % of adolescents experiencing depressive disorders by the age of 14 and 38 % suffering from anxiety disorders (Ersine et al., 2015; Ormel et al., 2015; Zhang et al., 2022). Depression is one of the most common mental disorders, triggered by the interplay of social, psychological, and biological factors, and is characterized by persistent depressed mood or loss of interest in daily activities (World Health Organization, 2023b). Due to

the emotional fluctuations and irritability common in adolescents, depression is often overlooked, especially when it manifests primarily through physical symptoms or behavioral issues (Hauenstein, 2003). Today, adolescent depression has become a serious mental health concern, with prevalence rates as high as 8 %–20 %, potentially leading to suicide, academic failure, poor peer relationships, substance abuse, and severe depression in adulthood (Muhil, 2015; Paul and Usha, 2021). This burden is particularly acute in low- and middle-income countries (Thapar et al., 2012). Anxiety disorders are the most common mental disorders in childhood, adolescence, and adulthood, affecting approximately one-quarter of adolescents, with a likelihood of comorbidity with depression and substance use disorders (Essau et al., 2018; Liu et al., 2024). The treatment rate for anxiety disorders is low, especially in low-income countries (Alonso et al., 2018).

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Existing Global Burden of Disease (GBD) studies indicate that in 2021, depression was the second leading cause of years lived with disability (YLDs), while anxiety ranked sixth ([The Lancet Psychiatry, 2024](#)). However, research on the trends in the global, regional, and national burden of depression and anxiety among adolescents and young people aged 10–24 years remains limited, particularly regarding its relationship with the Socio-Demographic Index (SDI). Therefore, this study aims to explore the following four questions through a comprehensive analysis of GBD 2021: (1) Analyze the trends in incidence, prevalence, and Disability-Adjusted Life Years (DALYs) of depression and anxiety among the global population aged 10–24, stratified by age, gender, region, and SDI; (2) Assess the transnational changes in the inequality of depression and anxiety burdens between 1990 and 2021, particularly in relation to SDI; (3) analyze the cross-national inequality of the burdens; (4) Provide evidence for relevant policies and interventions to improve the mental health of adolescents.

2. Methods

2.1. Study population and data collection

The GBD 2021 study estimated YLDs, years of life lost (YLLs), DALYs, and healthy life expectancy (HALE) for 371 diseases and injuries across 204 countries and territories (including subnational locations in 21 countries) from 1990 to 2021. Data were sourced from 100,983 inputs, including vital registration systems, verbal autopsies, censuses, household surveys, disease-specific registries, health service records, and government statistical resources identified through systematic searches and collaborator input ([Schumacher et al., 2024](#)). YLDs were calculated by multiplying the prevalence of each sequela by corresponding disability weights; YLLs were determined by multiplying cause-specific deaths by standard life expectancy at death. DALYs were the sum of YLDs and YLLs, while HALE was estimated using YLDs per capita and age-specific mortality rates. Counts and age-standardised rates were calculated globally, by super-region, region, country, and subnational location. All estimates incorporated uncertainty intervals (UIs) derived from the 2.5th and 97.5th percentiles of 500 draws, with uncertainty propagated at each estimation step. This analysis reports trends from 2010 to 2021, including the first two years of the COVID-19 pandemic. Comprehensive source details are available via the GBD 2021 Sources Tool ([Ferrari et al., 2024](#); [Global Burden of Disease \(GBD\), 2021](#); [Schumacher et al., 2024](#)).

The depression and anxiety disorders included in this study are classified under mental disorders in GBD 2021. Depressive and anxiety disorders were defined according to internationally recognized diagnostic criteria, including the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV, DSM-5) and the International Classification of Diseases (ICD-10, ICD-11) systems. Data sources included systematic reviews, population-based epidemiological surveys, and administrative health records. Estimates were synthesized using DisMod-MR 2.1, a Bayesian meta-regression tool designed to ensure consistency across data sources and adjust for variations in data quality. These methods have been extensively validated and published in peer-reviewed journals ([GBD 2019 Mental Disorders Collaborators, 2022](#); [Lin et al., 2021](#)).

Adolescence is defined by the World Health Organization (WHO) as the period from 10 to 19 years of age ([World Health Organization, 2023a](#)). To provide a more detailed description of adolescent development and an overall understanding of this life stage, we used age subgroups to define young adolescents (10–14 years), older adolescents (15–19 years), and young adults (20–24 years) ([Sawyer et al., 2018](#)). We conducted a retrospective ecological trend analysis using data from the GBD 2021 database to assess the burden of depressive and anxiety disorders among individuals aged 10–24 years from 1990 to 2021. Estimates of case counts, incidence rates, and DALYs were stratified by sex, age group (10–14, 15–19, 20–24 years), and SDI quintile, covering 21 regions and 204 countries and territories.

2.2. Socio-demographic index

Developed by GBD researchers, the SDI is a composite indicator of development status that is strongly correlated with health outcomes. It is calculated as the geometric mean of indices ranging from 0 to 1, including total fertility rate under the age of 25 (TFU25), mean education for those aged 15 and older (EDU15+), and lag-distributed income (LDI) per capita. As a composite, a location with an SDI of 0 represents a theoretical minimum level of development related to health, while an SDI of 1 indicates a theoretical maximum ([Global Burden of Disease, 2024](#)).

GBD 2021 calculated the SDI for each country, and we selected five SDI levels (high, high-middle, middle, low-middle, and low) to explore the relationship between the burden of depression and anxiety and socio-economic development among the population aged 10 to 24. For a more in-depth analysis of the relationship between SDI and the burdens of depression and anxiety, we classified the 204 countries and territories based on these five SDI levels and analyzed the associations of incidence rates, DALYs, and their corresponding average annual percent change (AAPC) with SDI using a linear smooth fitting curve model.

It is important to note that SDI differs fundamentally from individual-level socioeconomic status (SES). While SES typically reflects an individual's or household's education, income, and occupation, SDI is a macro-level indicator of national or regional development. Consequently, although this study analyzes mental disorder burden by SDI levels, it cannot assess changes among low-SES individuals during the pandemic ([Shavers, 2007](#); [Miech and Hauser, 2001](#); [Mueller and Parcel, 1981](#)).

2.3. Statistical analysis

We utilized Joinpoint trend analysis software (version 4.9.1.0) to analyze the global trends in incidence, prevalence, and DALY burdens of depression and anxiety among the population aged 10 to 24. This software fits segmented regression models based on Lerman's (1980) grid search algorithm and estimated *p*-values for permutation tests using Monte Carlo methods, while applying Bonferroni correction to maintain overall significance ([National Cancer Institute, 2022](#); [Kim et al., 2000](#)). Joinpoint regression is well-suited for analyzing long-term time-series data, as it objectively identifies statistically significant inflection points and estimates the AAPC for each segment. This method is more sensitive than conventional models in capturing stage-specific trend shifts—such as those before and after the COVID-19 pandemic—making it ideal for evaluating changes in disease burden over decades. The Annual Percent Change (APC) within each segment is calculated using the geometric mean, while the AAPC represents a weighted average of APCs across all intervals, reflecting the overall trend ([Clegg et al., 2009](#); [National Cancer Institute, 2022](#)). For example, an AAPC of 0.3 represents an annual growth rate of 0.3 %. We calculated the AAPC values for the years 1990–2021 as well as for four periods (1990–1997, 1998–2005, 2006–2013, 2014–2021), with trends reflected by the AAPC values and their 95 % confidence intervals (CIs).

Following WHO recommendations, we employed the slope index of inequality (SII) and concentration index (CI) to evaluate the distributional inequalities of depression and anxiety burdens ([World Health Organization, 2013a](#)). The SII is calculated by regressing DALY rates to an SDI-related relative position scale, defined as the midpoint of the cumulative population range ordered by SDI. The CI is computed as the area under the Lorenz concentration curve, indicating the cumulative distribution of disease burden relative to SDI ranking. If the Lorenz curve lies above the line of equality, the burden is concentrated in low-SDI countries, represented by a negative CI ([World Health Organization, 2013a](#); [Ordunez et al., 2019](#)). All statistical analyses were performed using RStudio (version 4.3.3) and Joinpoint.

3. Results

3.1. Global trends

Globally, depressive disorders incidence among adolescents and young adults increased between 1990 and 1997, between 1998 and 2005, and between 2014 and 2021, decreased between 2006 and 2013. Overall, the depressive disorder incidence in 2021 (3909.9 per 100,000 people [95 % UI: 2841.3–5171.1]) was higher than the incidence in 1990 (3085.5 per 100,000 people [2307.8–4105.6]; AAPC 0.97 [95 % CI: 0.75 to 1.19]). The prevalence and DALYs of depressive disorder decreased between 2006 and 2013, increased between 1990 and 1997, between 1998 and 2005, and between 2014 and 2021. Overall, the change was significant between 1990 and 2021 (Table 1). The Joinpoint regression analysis identified a substantial change in the incidence of depressive disorder in 2006, 2010 and 2019 (Fig. 1).

The total incidence of anxiety disorders increased during the periods 2006–2013 and 2014–2021, respectively. It decreased between 1998 and 2005 and remained unchanged between 1990 and 1997. Overall, the incidence of anxiety disorders increased from 708.0 per 100,000 (95 % UI: 511.5–939.3) in 1990 to 883.1 per 100,000 (636.9–1167.1) in 2021 (AAPC 0.84 [95 % CI: 0.48 to 1.21]).

The AAPC for the prevalence and DALYs of anxiety disorders decreased during 1990–1997 and 2006–2013, remained unchanged

Table 1

Global AAPCs in the prevalence, incidence, and DALYs for depressive disorder and anxiety disorders among adolescents aged 10–24 years.

	Incidence		Prevalence		DALYs	
	AAPC (95%CI)	p value	AAPC (95%CI)	p value	AAPC (95%CI)	p value
Depressive disorders						
1990–1997	0.14 (0.11 to 0.17)	<0.001	0.05 (0.01 to 0.08)	0.015	0.07 (0.04 to 0.11)	<0.001
1998–2005	0.36 (0.25 to 0.47)	<0.001	0.41 (0.34 to 0.49)	<0.001	0.43 (0.33 to 0.53)	<0.001
2006–2013	–0.65 (–0.71 to –0.58)	<0.001	–0.32 (–0.37 to –0.27)	<0.001	–0.46 (–0.52 to –0.40)	<0.001
2014–2021	4.67 (4.49 to 4.84)	<0.001	3.60 (3.41 to 3.79)	<0.001	4.15 (3.97 to 4.33)	<0.001
1990–2021	0.97 (0.75 to 1.19)	<0.001	0.80 (0.65 to 0.94)	<0.001	0.92 (0.66 to 1.18)	<0.001
Anxiety disorders						
1990–1997	–0.04 (–0.06 to –0.02)	<0.001	–0.13 (–0.15 to –0.10)	<0.001	–0.12 (–0.14 to –0.09)	<0.001
1998–2005	–0.28 (–0.32 to –0.24)	<0.001	–0.02 (–0.06 to 0.01)	0.191	0.01 (–0.06 to 0.07)	0.844
2006–2013	0.15 (0.12 to 0.18)	<0.001	–0.09 (–0.11 to –0.06)	<0.001	–0.09 (–0.11 to –0.06)	<0.001
2014–2021	4.04 (4.02 to 4.05)	<0.001	3.58 (3.52 to 3.63)	<0.001	3.54 (3.50 to 3.58)	<0.001
1990–2021	0.84 (0.48 to 1.21)	<0.001	0.77 (0.60 to 0.94)	<0.001	0.77 (0.60 to 0.93)	<0.001

Data in parentheses are 95 % confidence intervals for AAPCs.

DALYs = Disability-adjusted life-years; AAPC = average annual percentage change.

between 1998 and 2005, and increased during 2014–2021 (Table 1). Joinpoint regression analysis revealed a significant rise in the global incidence of anxiety disorders after 2019 (Fig. 1). In order to clarify the possible impact of COVID-19, we conducted a sensitivity analysis by excluding the data after 2019. The results revealed that between 1990 and 2019, anxiety - related indicators, including DALYs, incidence, and prevalence, exhibited a modest decline among 10–24 - year - olds. In contrast, the DALYs of depression remained comparatively stable, accompanied by a modest increase in the prevalence and a modest decrease in the incidence. However, since 2020 (following the COVID-19 outbreak), the burden and prevalence of anxiety and depression began to rise. Although the changes in 2020–2021 were not yet statistically significant when examined individually, the inclusion of these years in the full 1990–2021 trend analysis resulted in a statistically significant upward shift in all indicators. This suggests that the COVID-19 pandemic may have had a sustained negative impact on adolescent and youth mental health, altering the previous disease trend. (Supplemental Table S1).

3.2. Global trends by sex

There were global increases in depressive disorders incidence from 1990 to 2021 in females, with an AAPC of 0.93 (95 % CI: 0.69 to 1.17) and increases in incidence from 3932.7 (95 % UI: 2961.2–5221.7) per 100,000 people to 4875.1 (95 % UI: 3571.8–6378.8) per 100,000 people. Also, there were global increases in depressive disorders incidence from 1990 to 2021 in males (AAPC 1.07 [95 % CI: 0.82 to 1.32]) and an increase in incidence from 2265.2 per 100,000 people (95 % UI: 1684.2–3013.9) to 2991.7 per 100,000 people (95 % UI: 2155.5–3978.7), the DALYs of depressive disorders followed the same pattern in the sex (Table 2). There was an increase in the incidence of anxiety disorders in males from 567.5 per 100,000 people (95 % UI: 411.0–747.6) in 1990 to 712.1 per 100,000 people (515.2–935.3) in 2021 (AAPC 0.86 [95 % CI: 0.72 to 1.01]). Similar trends were seen in females, with an increase in incidence from 853.2 per 100,000 people (612.3–1138.4) to 1062.9 per 100,000 people (761.5–1423.1; AAPC 0.86 [95 % CI: 0.63 to 1.09]), the DALYs of anxiety disorders followed the same pattern in the sex (Table 2). Overall, in 2021, the incidence and DALYs of depressive and anxiety disorders in females were higher than those in males. However, from 1990 to 2021, the AAPC for incidence and DALYs of depression was greater in males than in females, while there was no significant difference in the AAPC of anxiety disorders between sexes.

3.3. Age groups trends

From 1990 to 2021, the incidence of depressive and anxiety disorders among the global population aged 10–24 showed an upward trend. The AAPC for depressive disorder incidence was 1.44 (95 % CI: 1.29 to 1.58) for the 10–14 age group, 1.05 (95 % CI: 0.83 to 1.27) for the 15–19 age group, and 0.81 (95 % CI: 0.74 to 0.87) for the 20–24 age group, with similar growth trends in DALYs for depression. The AAPC for anxiety disorder incidence was 0.78 (95 % CI: 0.55 to 1.02), 0.85 (95 % CI: 0.70 to 1.01), and 0.96 (95 % CI: 0.86 to 1.07) for the respective age groups, with DALYs also showing significant growth (see Table 2). The AAPC for depressive disorder incidence was highest in the 10–14 age group (1.44 %), with females (1.46 %) surpassing males (1.41 %); the AAPC for anxiety disorder incidence was highest in the 20–24 age group (0.96 %), with males (0.99 %) exceeding females (0.92 %) (Table 2 and Supplemental Table S2).

3.4. Global trends by SDI and regional trends

From 1990 to 2021, the AAPC in the incidence of depressive and anxiety disorders was highest in high SDI countries, at 1.92 (95 % CI: 1.70 to 2.14) and 1.12 (95 % CI: 0.67 to 1.57), respectively, while it was

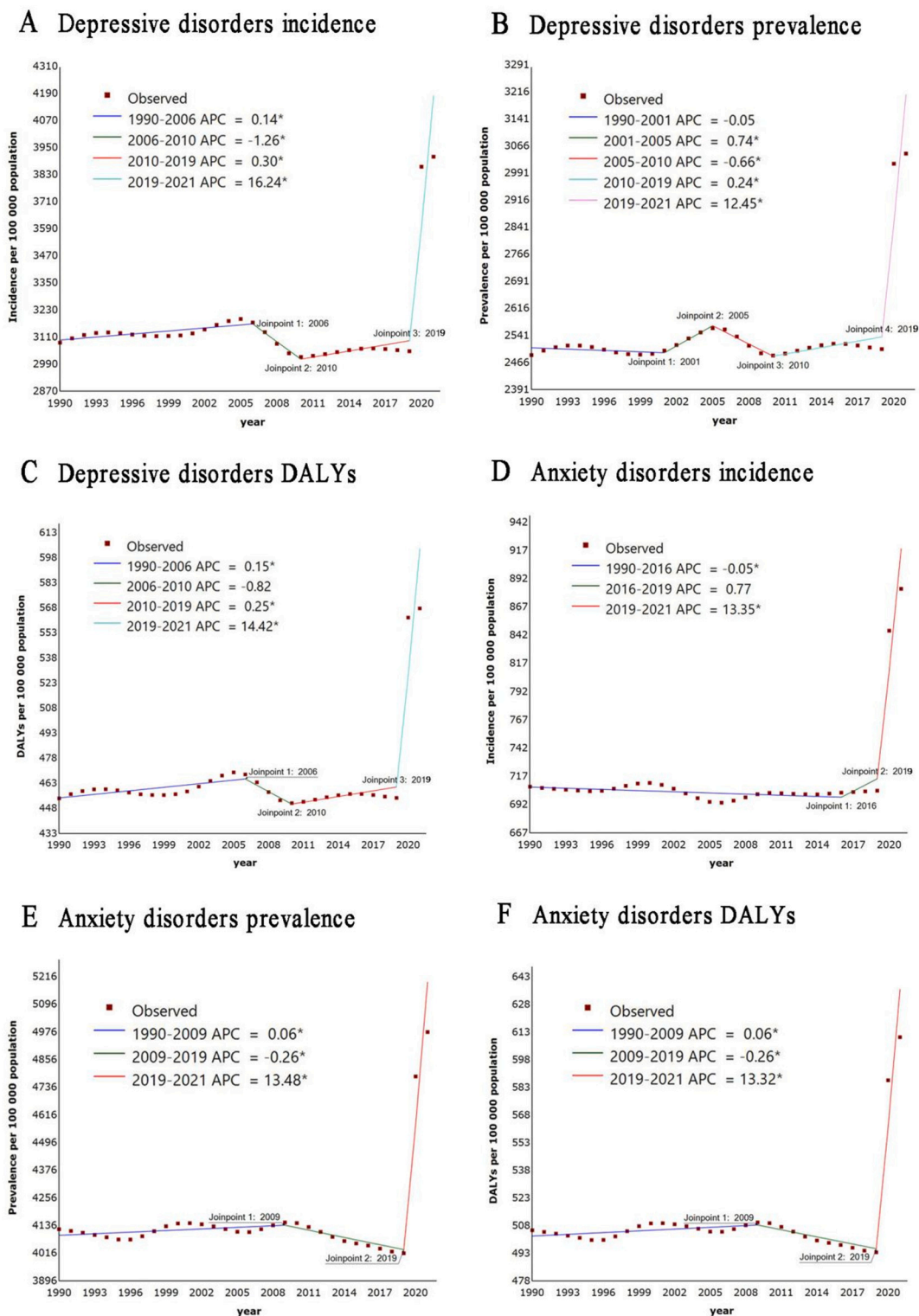


Fig. 1. Joinpoint regression analysis of global depressive disorders incidence (A), prevalence (B), and disability-adjusted life-years (DALYs) (C); and anxiety disorders incidence (D), prevalence (E), and DALYs (F) in adolescents and young adults aged 10–24 years from 1990 to 2021. APC = Annual percentage change; AAPC = average annual percentage change. * Indicates that the APC is significantly different from zero at the $\alpha = 0.05$ level Final.

lowest in low SDI countries. Similarly, the DALYs for both disorders had the highest AAPC in high SDI countries and the lowest in low SDI countries. The AAPC for depressive disorder incidence was highest in High-income North America, Central Latin America, and Andean Latin America, at 2.30 (95 % CI: 1.53 to 3.07), 2.00 (95 % CI: 1.72 to 2.28), and 1.68 (95 % CI: 1.22 to 2.14), respectively. Likewise, these three

regions also had the highest AAPC for DALYs.

For anxiety incidence, the top three AAPCs were in Andean Latin America, Tropical Latin America, and Central Latin America, at 1.91 (95 % CI: 1.82 to 2.00), 1.61 (95 % CI: 1.42 to 1.81), and 1.48 (95 % CI: 1.18 to 1.79), respectively. The regions with the highest DALYs AAPC were Tropical Latin America, Andean Latin America, and Central Latin

Table 2

Global and regional incidence, DALYs, and average annual percentage changes of depressive and anxiety disorders among 10–24-year-old adolescents from 1990 to 2021.

	Incidence					p value	DALYs					
	Cases (n), 1990	Incidence (per 100,000 people), 1990	Cases (n), 2021	Incidence (per 100,000 people), 2021	AAPC, 1990–2021		Cases (n), 1990	DALYs (per 100,000 people), 1990	Cases (n), 2021	DALYs (per 100,000 people), 2021	AAPC, 1990–2021	p value
Depressive disorder												
Global	47,739,042 (35706347–63,520,938)	3085.5 (2307.8–4105.6)	73,809,681 (53637291–97,618,586)	3909.9 (2841.3–5171.1)	0.97 (0.75 to 1.19)	<0.001	7,025,089 (4512560–10,171,661)	454.1 (291.7–657.4)	10,718,195 (6792136–15,633,301)	567.8 (359.8–828.1)	0.92 (0.66 to 1.18)	<0.001
Sex												
Male	17,806,864 (13239024–23,692,258)	2265.2 (1684.2–3013.9)	28,942,672 (20852925–38,491,309)	2991.7 (2155.5–3978.7)	1.07 (0.82 to 1.32)	<0.001	2,657,918 (1699507–3,883,716)	338.1 (216.2–494.1)	4,248,443 (2704223–6,258,555)	439.1 (279.5–646.9)	1.00 (0.74 to 1.27)	<0.001
Female	29,932,178 (22537815–39,742,261)	3932.7 (2961.2–5221.7)	44,867,009 (32872250–58,705,832)	4875.1 (3571.8–6378.8)	0.93 (0.69 to 1.17)	<0.001	4,367,171 (2813054–6,296,235)	573.8 (369.6–827.3)	6,469,752 (4113643–9,396,058)	703.0 (447.0–1020.9)	0.88 (0.59 to 1.17)	<0.001
Age group, years												
10–14	7,496,233 (4732105–10,869,040)	1399.4 (883.4–2029.0)	13,424,107 (8546539–19,698,774)	2013.7 (1282.0–2955.0)	1.44 (1.29 to 1.58)	<0.001	926,880 (553471–1,402,112)	173.0 (103.3–261.7)	1,629,554 (958147–2,495,215)	244.4 (143.7–374.3)	1.35 (1.25 to 1.45)	<0.001
15–19	17,929,058 (12623490–24,345,051)	3451.7 (2430.3–4686.9)	27,848,834 (19026208–37,536,128)	4463.1 (3049.2–6015.6)	1.05 (0.83 to 1.27)	<0.001	2,550,150 (1624371–3,808,419)	491.0 (312.7–733.2)	3,979,118 (2494528–6,048,542)	637.7 (399.8–969.3)	1.04 (0.88 to 1.20)	<0.001
20–24	22,313,750 (16624928–30,759,011)	4534.5 (3378.4–6250.7)	32,536,740 (23813252–46,626,264)	5448.6 (3987.8–7808.0)	0.81 (0.74 to 0.87)	<0.001	3,548,060 (2262644–5,269,857)	721.0 (459.8–1070.9)	5,109,523 (3217184–7,651,378)	855.6 (538.7–1281.3)	0.68 (0.44 to 0.93)	<0.001
Socio-demographic index												
High	8,074,817 (6279238–10,288,948)	4122.8 (3206.0–5253.3)	12,511,299 (9407431–15,996,313)	6741.9 (5069.3–8619.9)	1.92 (1.70 to 2.14)	<0.001	1,200,974 (783360–1,723,403)	613.2 (400.0–879.9)	1,785,617 (1165099–2,617,760)	962.2 (627.8–1410.6)	1.75 (1.57 to 1.92)	<0.001
High-middle	8,501,874 (6542527–11,043,421)	2995.9 (2305.5–3891.5)	7,977,472 (5895769–10,454,887)	3531.7 (2610.1–4628.5)	0.64 (0.44 to 0.84)	<0.001	1,266,357 (824677–1,827,131)	446.2 (290.6–643.8)	1,170,484 (734971–1,738,150)	518.2 (325.4–769.5)	0.57 (0.36 to 0.78)	<0.001
Low	5,221,225 (3673134–7,084,885)	3354.2 (2359.7–4551.5)	14,398,342 (10007286–19,592,572)	3897.6 (2708.9–5303.7)	0.51 (0.31 to 0.71)	<0.001	759,817 (472159–1,116,982)	488.1 (303.3–717.6)	2,094,964 (1268940–3,087,420)	567.1 (343.5–835.8)	0.51 (0.33 to 0.69)	<0.001
Low-middle	11,291,507 (8331776–15,362,090)	3121.7 (2303.4–4247.0)	21,250,480 (15232171–28,609,514)	3844.5 (2755.7–5175.9)	0.89 (0.66 to 1.12)	<0.001	1,634,993 (1038201–2,408,604)	452.0 (287.0–665.9)	3,084,347 (1946037–4,492,344)	558.0 (352.1–812.7)	0.89 (0.68 to 1.10)	<0.001
Middle	14,609,672 (10911915–19,541,306)	2662.0 (1988.2–3560.6)	17,617,056 (13056072–23,397,245)	3187.2 (2362.0–4232.9)	0.73 (0.34 to 1.12)	<0.001	2,157,055 (1383168–3,136,955)	393.0 (252.0–571.6)	2,574,805 (1643530–3,783,562)	465.8 (297.3–684.5)	0.67 (0.28 to 1.07)	0.001
Region												
Andean Latin America	280,214 (199824–380,608)	2276.2 (1623.2–3091.8)	604,570 (411299–867,257)	3501.9 (2382.4–5023.4)	1.68 (1.22 to 2.14)	<0.001	41,530 (25975–60,896)	337.4 (211.0–494.7)	87,264 (53200–132,552)	505.5 (308.1–767.8)	1.57 (1.17 to 1.97)	<0.001
Australasia	323,548 (241386–416,322)	6725.0 (5017.2–8653.3)	428,706 (299787–584,185)	7472.4 (5225.3–10,182.5)	0.40 (0.13 to 0.68)	0.004	46,192 (29725–67,373)	960.1 (617.8–1400.4)	60,893 (38457–92,590)	1061.4 (670.3–1613.9)	0.39 (0.10 to 0.68)	0.007
Caribbean	397,964 (283916–539,008)	3726.7 (2658.7–5047.5)	471,217 (325337–670,524)	4159.9 (2872.1–5919.4)	0.50 (0.34 to 0.67)	<0.001	56,923 (35793–84,202)	533.0 (335.2–788.5)	66,837 (40575–100,480)	590.0 (358.2–887.0)	0.45 (0.28 to 0.63)	<0.001
Central Asia	544,999 (385807–733,041)	2748.2 (1945.5–3696.4)	776,936 (540712–1,085,239)	3511.2 (2443.6–4904.5)	0.98 (0.88 to 1.09)	<0.001	81,623 (51221–121,701)	411.6 (258.3–613.7)	114,009 (71936–169,649)	515.2 (325.1–766.7)	0.90 (0.80 to 1.00)	<0.001
Central Europe	610,705 (446867–820,500)	2091.6 (1530.5–2810.1)	515,528 (370550–709,804)	2842.5 (2043.2–3913.7)	1.12 (1.00 to 1.25)	<0.001	94,145 (60194–136,808)	322.4 (206.2–468.5)	77,220 (49111–112,479)	425.8 (270.8–620.2)	1.01 (0.80 to 1.22)	<0.001
Central Latin America	1,254,360 (926977–1,684,641)	2311.9 (1708.5–3105.0)	2,469,993 (1814447–3,336,508)	3798.1 (2790.0–5130.5)	2.00 (1.72 to 2.28)	<0.001	183,906 (117587–270,293)	339.0 (216.7–498.2)	350,939 (223149–518,516)	539.6 (343.1–797.3)	1.86 (1.60 to 2.11)	<0.001
Central Sub-Saharan Africa	964,729 (651094–1,367,890)	5573.9 (3761.8–7903.3)	2,703,093 (1773498–3,895,034)	6014.4 (3946.0–8666.4)	0.37 (0.21 to 0.54)	<0.001	134,054 (81237–201,919)	774.5 (469.4–1166.6)	376,502 (222278–571,808)	837.7 (494.6–1272.3)	0.38 (0.25 to 0.51)	<0.001
East Asia	8,670,892 (6558652–11,470,919)	2329.8 (1762.2–3082.1)	3,397,308 (2547414–4,427,705)	1398.0 (1048.3–1822.0)	–1.67 (–1.89 to –1.44)	<0.001	1,303,974 (831087–1,892,288)	350.4 (223.3–508.4)	535,009 (345137–776,864)	220.2 (142.0–319.7)	–1.51 (–1.77 to –1.26)	<0.001
Eastern Europe	1,288,899 (948264–1,731,985)	2729.1 (2007.8–3667.2)	1,254,429 (905921–1,697,740)	3801.9 (2745.6–5145.5)	1.09 (0.96 to 1.22)	<0.001	191,172 (123695–278,442)	404.8 (261.9–589.6)	179,232 (115357–263,048)	543.2 (349.6–797.2)	0.97 (0.88 to 1.05)	<0.001
Eastern Sub-Saharan Africa	2,254,637 (1612156–3,034,495)	3634.5 (2598.8–4891.7)	6,321,964 (4356705–8,663,916)	4346.9 (2995.6–5957.2)	0.65 (0.55 to 0.75)	<0.001	332,560 (208539–490,411)	536.1 (336.2–790.6)	926,424 (568457–1,357,235)	637.0 (390.9–933.2)	0.62 (0.55 to 0.70)	<0.001

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Table 2 (continued)

	Incidence					DALYs						
	Cases (n), 1990	Incidence (per 100,000 people), 1990	Cases (n), 2021	Incidence (per 100,000 people), 2021	AAPC, 1990–2021	p value	Cases (n), 1990	DALYs (per 100,000 people), 1990	Cases (n), 2021	DALYs (per 100,000 people), 2021	AAPC, 1990–2021	p value
High-income	1,119,774	2658.0	976,061	3739.0	1.18 (1.07to	<0.001	167,392	397.3	142,956	547.6	1.10 (1.04 to	<0.001
Asia Pacific	(843789–1,436,710)	(2002.9–3410.3)	(719517–1,295,670)	(2756.2–4963.3)	1.28)		(107711–240,653)	(255.7–571.2)	(90977–214,074)	(348.5–820.1)	1.16)	
High-income	3,137,300	5128.6	7,045,511	9884.4	2.30 (1.53to	<0.001	467,202	763.7	988,790	1387.2	2.26 (1.71 to	<0.001
North America	(2404609–4,037,578)	(3930.8–6600.2)	(5513507–8,913,319)	(7735.1–12,504.8)	3.07)		(307931–674,946)	(503.4–1103.3)	(667742–1,435,479)	(936.8–2013.9)	2.81)	
North Africa and Middle East	5,198,201	4773.1	9,842,270	6064.4	1.10 (0.79to	<0.001	750,117	688.8	1,409,383	868.4	0.97 (0.42 to	<0.001
	(3691130–6,997,012)	(3389.3–6424.8)	(6780495–13,664,266)	(4177.8–8419.3)	1.41)		(463752–1,119,163)	(425.8–1027.6)	(837694–2,147,155)	(516.1–1323.0)	1.51)	
Oceania	61,134 (42795–86,611)	2922.4 (2045.8–4140.3)	126,907 (84413–181,242)	3147.2 (2093.4–4494.7)	0.16 (0.12to	<0.001	8852 (5529–13,314)	423.2 (264.3–636.5)	18,385 (10699–28,632)	455.9 (265.3–710.1)	0.17 (0.12 to	<0.001
					0.19)						0.21)	
South Asia	9,604,922	2871.7	18,281,089	3476.3	0.86 (0.74to	<0.001	1,390,864	415.8	2,668,685	507.5	0.86 (0.74 to	<0.001
	(7145543–13,003,646)	(2136.4–3887.8)	(13436020–24,475,212)	(2555.0–4654.2)	0.99)		(884030–2,043,039)	(264.3–610.8)	(1677332–3,868,540)	(319.0–735.6)	0.98)	
Southeast Asia	3,270,260	2204.2	5,015,565	2932.9	0.93 (0.82to	<0.001	492,271	331.8	744,160	435.2	0.88 (0.78 to	<0.001
	(2396535–4,331,212)	(1615.3–2919.3)	(3584888–6,866,744)	(2096.3–4015.4)	1.04)		(315220–718,391)	(212.5–484.2)	(468701–1,091,144)	(274.1–638.1)	0.99)	
Southern Latin America	591,413	4468.0	919,891	5998.1	1.37 (1.09to	<0.001	84,884	641.3	130,746	852.5	1.30 (1.04 to	<0.001
	(444295–752,580)	(3356.5–5685.5)	(659060–1,227,120)	(4297.3–8001.3)	1.66)		(54632–122,901)	(412.7–928.5)	(83256–190,564)	(542.9–1242.6)	1.55)	
Southern Sub-Saharan Africa	533,142	3120.8	1,009,599	4628.1	1.47 (1.39to	<0.001	79,169	463.4	144,888	664.2	1.34 (1.27 to	<0.001
	(398426–709,916)	(2332.2–4155.5)	(735760–1,343,803)	(3372.8–6160.1)	1.54)		(50389–114,472)	(295.0–670.1)	(90671–211,209)	(415.6–968.2)	1.42)	
Tropical Latin America	1,681,640	3513.7	2,307,653	4562.7	1.04 (0.50to	<0.001	236,248	493.6	321,988	636.6	1.01 (0.49 to	<0.001
	(1260047–2,235,887)	(2632.8–4671.8)	(1729764–3,097,352)	(3420.1–6124.1)	1.57)		(151562–343,527)	(316.7–717.8)	(206527–477,624)	(408.3–944.4)	1.54)	
Western Europe	4,212,258	5124.4	4,527,472	6281.6	0.83 (0.56to	<0.001	623,828	758.9	654,187	907.6	0.75 (0.51 to	<0.001
	(3334047–5,270,388)	(4056.0–6411.6)	(3250728–6,038,278)	(4510.2–8377.7)	1.10)		(415298–893,628)	(505.2–1087.1)	(418715–971,956)	(580.9–1348.5)	0.98)	
Western Sub-Saharan Africa	1,738,051	2904.3	4,813,917	2983.2	0.26	0.072	258,183	431.4	719,696	446.0	0.26 (0.00 to	0.048
	(1249712–2,375,163)	(2088.3–3968.9)	(3444264–6,600,077)	(2134.4–4090.1)	(–0.02to 0.54)		(161470–376,842)	(269.8–629.7)	(447969–1,054,866)	(277.6–653.7)	0.52)	
Anxiety disorders												
Global	10,954,492	708.0	16,670,880	883.1	0.84 (0.48to	<0.001	7,822,522	505.6	11,523,716	610.4	0.77 (0.60 to	<0.001
	(7914603–14,532,409)	(511.5–939.3)	(12024089–22,032,684)	(636.9–1167.1)	1.21)		(4925459–11,445,029)	(318.3–739.7)	(7273258–16,910,890)	(385.3–895.8)	0.93)	
Sex												
Male	4,460,963	567.5	6,889,038	712.1	0.86 (0.72to	<0.001	3,050,246	388.0	4,527,687	468.0	0.75 (0.59 to	<0.001
	(3231120–5,877,137)	(411.0–747.6)	(4984436–9,048,520)	(515.2–935.3)	1.01)		(1926974–4,514,805)	(245.1–574.3)	(2839711–6,676,433)	(293.5–690.1)	0.92)	
Female	6,493,530	853.2	9,781,842	1062.9	0.86 (0.63to	<0.001	4,772,275	627.0	6,996,029	760.2	0.75 (0.52 to	<0.001
	(4660298–8,664,696)	(612.3–1138.4)	(7008114–13,097,386)	(761.5–1423.1)	1.09)		(3013165–6,962,633)	(395.9–914.8)	(4414313–10,246,474)	(479.6–1113.4)	0.99)	
Age group, years												
10–14	4,076,313	761.0	6,166,868	925.1	0.78 (0.55to	<0.001	2,238,635	417.9	3,358,105	503.7	0.73 (0.46 to	<0.001
	(3108602–5,308,851)	(580.3–991.0)	(4693642–8,089,046)	(704.1–1213.4)	1.02)		(1350319–3,327,223)	(252.1–621.1)	(2013418–5,072,153)	(302.0–760.9)	1.01)	
15–19	3,620,228	697.0	5,430,290	870.3	0.85 (0.70to	<0.001	2,831,156	545.1	4,054,128	649.7	0.73 (0.59 to	<0.001
	(2331461–5,039,146)	(448.9–970.1)	(3454697–7,622,176)	(553.7–1221.5)	1.01)		(1787407–4,215,876)	(344.1–811.6)	(2545178–6,004,786)	(407.9–962.3)	0.86)	
20–24	3,257,952	662.1	5,073,722	849.6	0.96 (0.86 to	<0.001	2,752,731	559.4	4,111,484	688.5	0.82 (0.70 to	<0.001
	(1813694–4,806,943)	(368.6–976.8)	(2897232–7,425,000)	(485.2–1243.4)	1.07)		(1738657–4,102,815)	(353.3–833.8)	(2580955–6,110,620)	(432.2–1023.3)	0.95)	
Socio-demographic index												
High	1,742,935	889.9	2,274,834	1225.8	1.12 (0.67 to	<0.001	1,258,744	642.7	1,584,215	853.7	0.99 (0.67 to	<0.001
	(1268323–2,295,511)	(647.6–1172.0)	(1640893–2,991,723)	(884.2–1612.1)	1.57)		(786738–1,850,574)	(401.7–944.9)	(1000566–2,303,100)	(539.2–1241.1)	1.30)	
High-middle	2,049,549	722.2	2,084,464	922.8	0.84 (0.76 to	<0.001	1,574,262	554.7	1,558,092	689.8	0.76 (0.69 to	<0.001
	(1496240–2,706,818)	(527.2–953.8)	(1533623–2,780,787)	(679.0–1231.1)	0.92)		(1013841–2,314,136)	(357.3–815.5)	(976353–2,307,595)	(432.2–1021.6)	0.82)	
Low	1,022,114	656.6	2,888,375	781.9	0.67 (0.49 to	<0.001	659,107	423.4	1,905,522	515.8	0.77 (0.69 to	<0.001
	(727873–1,379,168)	(467.6–886.0)	(2061186–3,865,353)	(558.0–1046.3)	0.84)		(411570–970,693)	(264.4–623.6)	(1156774–2,829,340)	(313.1–765.9)	0.84)	
Low-middle	2,282,198	630.9	4,390,669	794.3	0.89 (0.71 to	<0.001	1,468,968	406.1	2,893,764	523.5	0.94 (0.67 to	<0.001
	(1626631–3,065,961)	(449.7–847.6)	(3129601–5,882,210)	(566.2–1064.2)	1.07)		(922131–2,155,676)	(254.9–596.0)	(1835906–4,230,364)	(332.1–765.3)	1.21)	

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Table 2 (continued)

	Incidence					DALYs						
	Cases (n), 1990	Incidence (per 100,000 people), 1990	Cases (n), 2021	Incidence (per 100,000 people), 2021	AAPC, 1990–2021	p value	Cases (n), 1990	DALYs (per 100,000 people), 1990	Cases (n), 2021	DALYs (per 100,000 people), 2021	AAPC, 1990–2021	p value
Middle Region	3,847,857 (2777918–5,064,049)	701.1 (506.2–922.7)	5,018,674 (3600932–6,634,076)	908.0 (651.5–1200.2)	0.98 (0.76 to 1.21)	<0.001	2,854,435 (1829943–4,168,482)	520.1 (333.4–759.5)	3,572,197 (2263863–5,228,392)	646.3 (409.6–945.9)	0.81 (0.75 to 0.88)	<0.001
Andean Latin America	117,924 (80371–168,535)	957.9 (652.9–1369.0)	250,939 (160224–373,069)	1453.5 (928.1–2160.9)	1.91 (1.82 to 2.00)	<0.001	81,952 (49863–123,218)	665.7 (405.1–1000.9)	176,030 (103403–270,836)	1019.6 (598.9–1568.8)	1.62 (1.31 to 1.92)	<0.001
Australasia	57,849 (38987–80,867)	1202.4 (810.4–1680.8)	78,421 (51001–111,380)	1366.9 (889.0–1941.4)	0.49 (0.33 to 0.65)	<0.001	41,624 (25840–61,459)	865.2 (537.1–1277.4)	53,007 (31679–78,764)	923.9 (552.2–1372.9)	0.29 (0.19 to 0.39)	<0.001
Caribbean	85,407 (59211–120,108)	799.8 (554.5–1124.7)	116,190 (78428–170,008)	1025.7 (692.4–1500.8)	0.96 (0.92 to 1.00)	<0.001	60,982 (36466–92,750)	571.1 (341.5–868.5)	82,589 (48852–129,329)	729.1 (431.3–1141.7)	0.92 (0.86 to 0.98)	<0.001
Central Asia	85,538 (59230–120,360)	431.3 (298.7–606.9)	123,139 (84104–177,976)	556.5 (380.1–804.3)	0.93 (0.69 to 1.17)	<0.001	57,509 (34304–87,762)	290.0 (173.0–442.5)	82,852 (49892–125,285)	374.4 (225.5–566.2)	0.98 (0.83 to 1.13)	<0.001
Central Europe	181,810 (128407–247,569)	622.7 (439.8–847.9)	163,108 (113710–224,369)	899.4 (627.0–1237.1)	1.29 (1.24 to 1.34)	<0.001	126,202 (77354–188,999)	432.2 (264.9–647.3)	113,205 (69640–168,854)	624.2 (384.0–931.0)	1.27 (1.12 to 1.43)	<0.001
Central Latin America	374,947 (267445–500,549)	691.1 (492.9–922.6)	671,585 (470453–912,400)	1032.7 (723.4–1403.0)	1.48 (1.18 to 1.79)	<0.001	266,160 (166189–397,368)	490.6 (306.3–732.4)	475,927 (295684–704,186)	731.8 (454.7–1082.8)	1.50 (0.75 to 2.25)	<0.001
Central Sub-Saharan Africa	125,262 (86196–175,924)	723.7 (498.0–1016.4)	368,600 (247670–542,382)	820.1 (551.1–1206.8)	0.53 (0.39 to 0.67)	<0.001	83,395 (50992–125,853)	481.8 (294.6–727.1)	246,731 (147587–385,307)	549.0 (328.4–857.3)	0.56 (0.41 to 0.71)	<0.001
East Asia	2,438,390 (1752796–3,202,210)	655.2 (471.0–860.4)	1,701,656 (1265938–2,226,837)	700.2 (520.9–916.4)	0.10 (0.01 to 0.19)	0.024	1,996,747 (1297881–2,902,881)	536.5 (348.7–780.0)	1,333,188 (858073–1,973,011)	548.6 (353.1–811.9)	–0.04 (–0.15to0.07)	0.501
Eastern Europe	315,623 (227584–415,460)	668.3 (481.9–879.7)	323,485 (234531–426,388)	980.4 (710.8–1292.3)	1.28 (1.26 to 1.30)	<0.001	217,432 (140543–320,153)	460.4 (297.6–677.9)	219,467 (140098–322,756)	665.2 (424.6–978.2)	1.23 (1.18 to 1.27)	<0.001
Eastern Sub-Saharan Africa	452,491 (317308–618,440)	729.4 (511.5–996.9)	1,293,468 (888587–1,768,224)	889.4 (611.0–1215.8)	0.70 (0.64 to 0.76)	<0.001	282,463 (176004–413,196)	455.3 (283.7–666.1)	820,303 (500697–1,217,616)	564.0 (344.3–837.2)	0.75 (0.66 to 0.84)	<0.001
High-income Asia Pacific	258,594 (185623–338,850)	613.8 (440.6–804.3)	193,764 (138330–258,348)	742.3 (529.9–989.7)	0.58 (0.53 to 0.62)	<0.001	190,850 (120185–283,941)	453.0 (285.3–674.0)	142,181 (89113–208,753)	544.7 (341.4–799.7)	0.59 (0.51 to 0.66)	<0.001
High-income North America	588,367 (415672–777,801)	961.8 (679.5–1271.5)	984,972 (676344–1,299,587)	1381.9 (948.9–1823.2)	1.33 (0.61 to 2.05)	<0.001	357,811 (234353–510,227)	584.9 (383.1–834.1)	589,457 (384993–854,035)	827.0 (540.1–1198.2)	1.29 (0.72 to 1.86)	<0.001
North Africa and Middle East	1,113,941 (834250–1,469,423)	1022.8 (766.0–1349.3)	2,061,402 (1486147–2,797,949)	1270.1 (915.7–1724.0)	0.91 (0.57 to 1.24)	<0.001	878,967 (550238–1,276,172)	807.1 (505.2–1171.8)	1,623,594 (1002404–2,401,134)	1000.4 (617.6–1479.5)	0.88 (0.78 to 0.98)	<0.001
Oceania	15,331 (10942–21,179)	732.9 (523.1–1012.4)	33,796 (22418–49,582)	838.1 (555.9–1229.6)	0.34 (0.21 to 0.47)	<0.001	11,285 (6814–17,269)	539.5 (325.7–825.5)	25,121 (13988–40,143)	623.0 (346.9–995.5)	0.36 (0.31 to 0.41)	<0.001
South Asia	1,792,408 (1259077–2,390,416)	535.9 (376.4–714.7)	3,478,268 (2378342–4,619,814)	661.4 (452.3–878.5)	0.82 (0.46 to 1.18)	<0.001	1,034,132 (658731–1,505,541)	309.2 (196.9–450.1)	2,071,746 (1315567–2,971,457)	394.0 (250.2–565.1)	0.95 (0.61 to 1.28)	<0.001
Southeast Asia	929,074 (662090–1,234,337)	626.2 (446.3–832.0)	1,454,024 (1027457–1,970,472)	850.3 (600.8–1152.3)	1.01 (0.98 to 1.03)	<0.001	619,650 (388124–916,651)	417.6 (261.6–617.8)	985,255 (637138–1,469,551)	576.1 (372.6–859.3)	1.05 (1.02 to 1.09)	<0.001
Southern Latin America	131,614 (97480–176,935)	994.3 (736.4–1336.7)	202,068 (133411–289,893)	1317.6 (869.9–1890.2)	1.14 (0.79 to 1.49)	<0.001	98,891 (61992–145,740)	747.1 (468.3–1101.0)	151,569 (87572–231,566)	988.3 (571.0–1509.9)	1.15 (0.79 to 1.50)	<0.001
Southern Sub-Saharan Africa	121,172 (86800–160,473)	709.3 (508.1–939.3)	215,812 (152930–289,589)	989.3 (701.0–1327.5)	1.19 (1.13 to 1.24)	<0.001	82,372 (52311–120,810)	482.2 (306.2–707.2)	147,177 (93426–212,910)	674.7 (428.3–976.0)	1.22 (0.94 to 1.51)	<0.001
Tropical Latin America	479,981 (344154–629,409)	1002.9 (719.1–1315.1)	802,553 (554623–1,082,144)	1586.8 (1096.6–2139.6)	1.61 (1.42 to 1.81)	<0.001	337,618 (219721–480,176)	705.4 (459.1–1003.3)	572,225 (371419–824,730)	1131.4 (734.4–1630.7)	1.68 (1.55 to 1.81)	<0.001
Western Europe	915,845 (657805–1,217,657)	1114.2 (800.2–1481.3)	1,050,410 (756578–1,393,582)	1457.4 (1049.7–1933.5)	1.10 (0.97 to 1.23)	<0.001	732,724 (450134–1,075,628)	891.4 (547.6–1308.5)	825,862 (516691–1,215,370)	1145.8 (716.9–1686.2)	1.03 (0.96 to 1.10)	<0.001
Western Sub-Saharan Africa	372,923 (271667–490,756)	623.2 (454.0–820.1)	1,103,219 (805535–1,446,126)	683.7 (499.2–896.2)	0.40 (0.23 to 0.58)	<0.001	263,755 (163791–392,060)	440.7 (273.7–655.1)	786,230 (487065–1,172,646)	487.2 (301.8–726.7)	0.44 (0.23 to 0.64)	<0.001

Data in parentheses are 95 % uncertainty intervals for cases, incidence, and DALYs, and 95 % confidence intervals for AAPCs.

DALYs = Disability-adjusted life-years; AAPC = average annual percentage change.

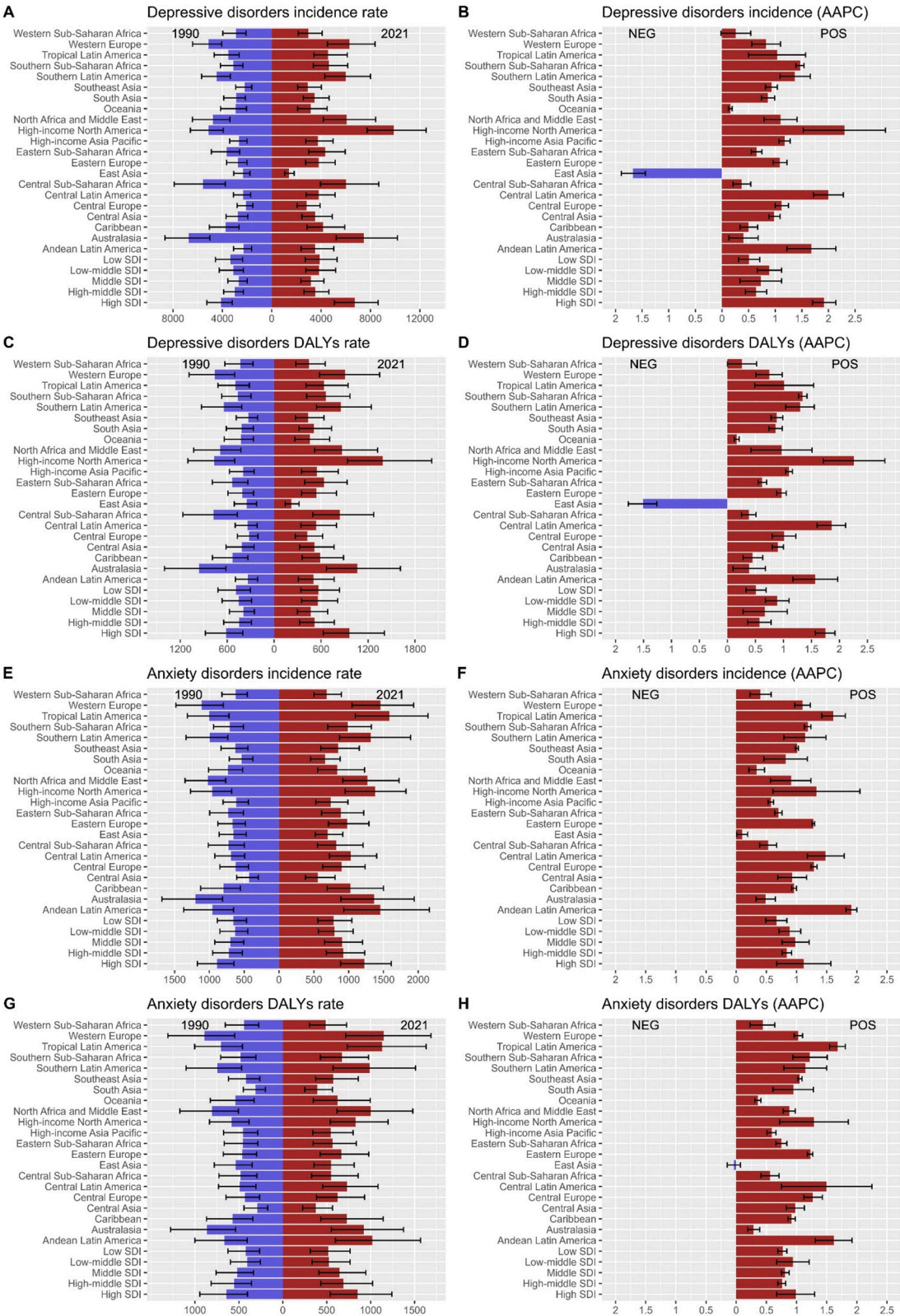


Fig. 2. The changes in incidence, DALYs, and AAPC of depressive and anxiety disorders across various regions between 1990 and 2021. The bar charts, along with error bars, display the central estimates and their 95 % uncertainty intervals for incidence rates (A and E) and DALY rates (C and G). The AAPC for incidence rates (B and F) and DALY rates (D and H) are also shown, with error bars representing their 95 % confidence intervals. The positive and negative signs of AAPC are represented by NEG and POS respectively. All data pertain to individuals aged 10 to 24 years.

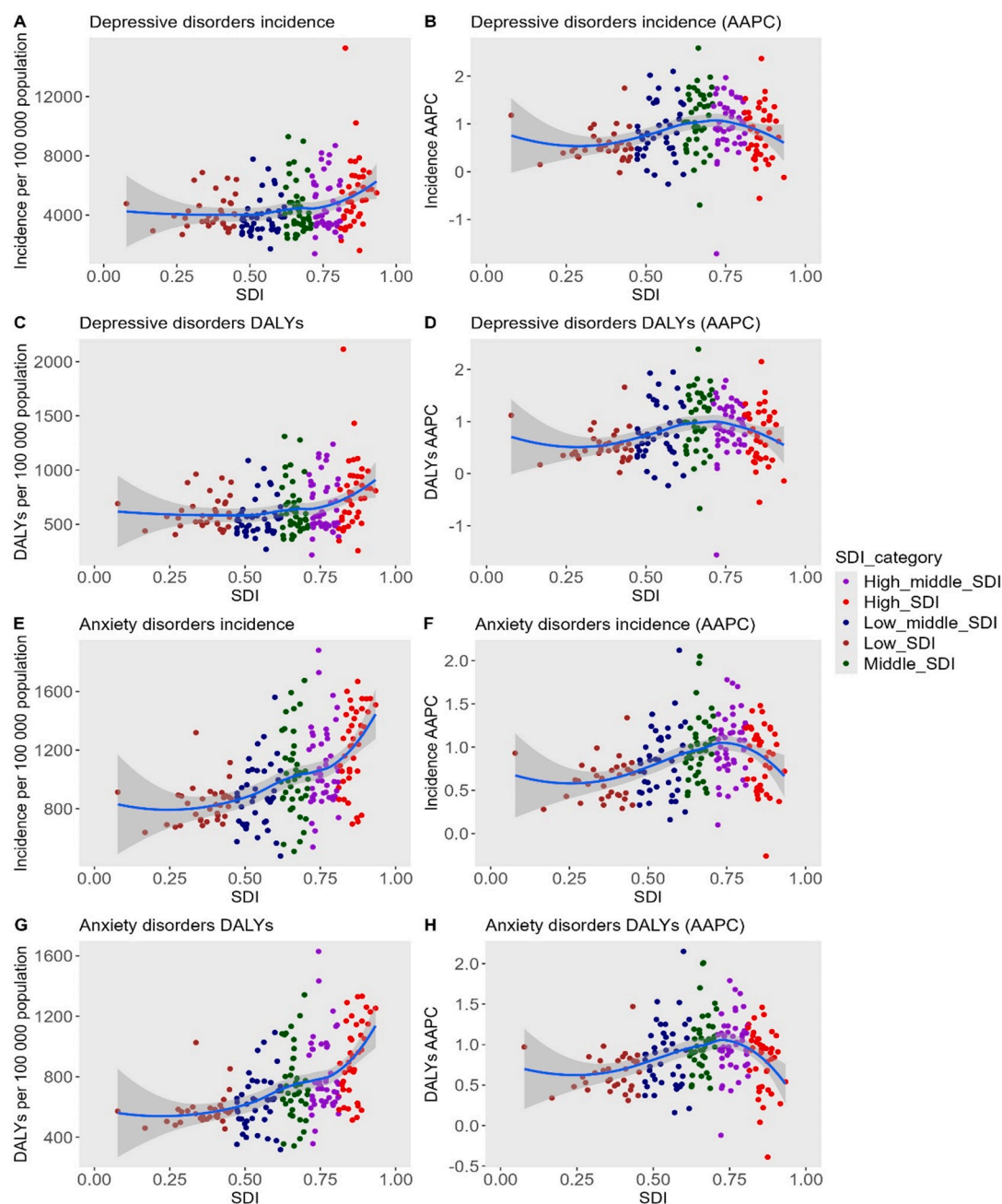


Fig. 3. Overall trends in the incidence and disease burden of depression and anxiety among individuals aged 10–24 by SDI (Overall). Trends in incidence and DALYs of depressive and anxiety disorders and the SDI (A, C, E, G), as well as the AAPC in these metrics across different SDI levels (B, D, F, H). The blue lines represent the fitted curves, with the shaded gray areas indicating the 95 % confidence intervals. The figures demonstrate trends in the incidence and burden of depressive and anxiety disorders as SDI increases, along with their temporal changes as reflected in AAPC values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

America. However, East Asia exhibited decline in both the incidence and DALYs AAPC for depression, and this region had the slowest growth in anxiety incidence and DALYs AAPC. (Table 2 and Fig. 2).

3.5. Countries and territories trends

Overall, the incidence and DALYs of depressive and anxiety disorders showed an upward trend with increasing SDI, particularly higher in high SDI regions. However, the AAPC for depression incidence and DALYs displayed a slight downward trend in middle-high SDI regions, while the AAPC for anxiety disorder incidence and DALYs showed a decreasing trend in high SDI regions, indicating that the rate of increase in the

incidence of these disorders is slowing down, as is the growth rate of the overall health burden caused by depression and anxiety (Fig. 3). Locally, although high SDI countries generally have higher incidence and burden of both disorders, significant differences exist between countries within this group, with some high SDI countries exhibiting lower burdens, while some low SDI countries display relatively high incidence or burden, as shown by similar AAPCs for both disorders (Supplemental Fig. 1). The countries and territories with the highest AAPC for the incidence and DALYs of depressive disorders are Mexico and the United States of America, while China has the lowest AAPC for both incidence and DALYs. In contrast, Bolivia and Mexico lead in the incidence and DALYs of anxiety disorders, with the lowest AAPC observed in Taiwan

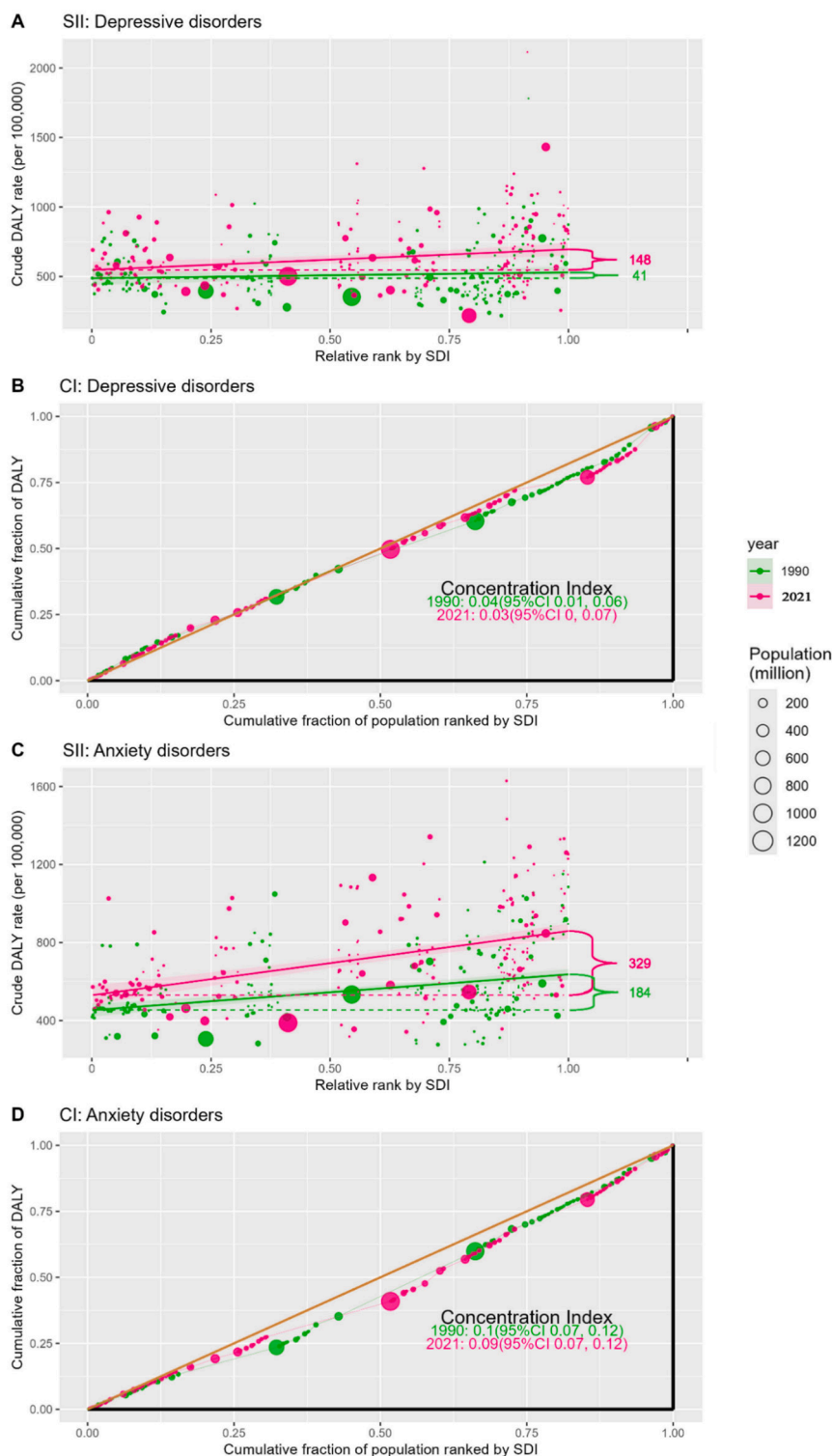


Fig. 4. Absolute SDI-related health inequality in the burden of depressive and anxiety, shown using regression lines for 1990 and 2021. Dots represent countries and territories, with sizes indicating population (A and C). Relative SDI-related health inequality in the burden of depressive and anxiety, shown using concentration curves for 1990 and 2021. Dots represent countries and territories, with sizes indicating population (B and D).

(Province of China). (Supplemental Table S3).

3.6. Health inequality analysis

In terms of absolute inequality, the crude DALYs rate gap between high and low SDI countries significantly increased from 1990 to 2021, specifically as follows: depressive disorders increased from 41 in 1990 to

148 in 2021, and anxiety disorders rose from 184 to 329 during the same period. This indicates an expanding gap in the burden of depression and anxiety between high and low SDI countries, particularly with a significant increase in the burden of these mental disorders in high SDI countries. Conversely, regarding relative inequality, the gap in crude DALYs rate between high and low SDI countries decreased from 1990 to 2021, with depressive disorders decreasing from 0.04 to 0.03, and

anxiety disorders from 0.10 to 0.09. This suggests that while the burden in high SDI countries has increased, the burden in low SDI countries has also risen, thereby partially narrowing the relative gap (Fig. 4). In summary, the increase in absolute inequality may reflect a continuing deterioration of the health burden in high SDI countries, while the reduction in relative inequality may indicate that low SDI countries are experiencing a faster increase in mental health burdens, leading to a narrowing of the relative gap with high SDI countries.

4. Discussion

Based on GBD 2021 data, this study analyzed the incidence, prevalence, DALYs, and AAPC of depression and anxiety among the global population aged 10–24, and explored their distribution in relation to age groups, gender, regions, SDI, and across 204 countries and territories, along with an inequality analysis related to SDI. The strengths of this study include a long observation period (1990–2021), extensive data coverage, and a detailed presentation of the distribution and inequality of depression and anxiety among the 10–24 age group through overall and local trend analyses combined with SII and CI metrics.

The findings reveal an overall increasing trend in the incidence, prevalence, and AAPC of depression and anxiety over the 32 years, with a notable decline in the AAPC for these metrics during the period from 2006 to 2013. This decline may be associated with heightened global awareness of depression and the implementation of related policies, such as the Mental Health Gap Action Programme in January 2008 and the European Pact for Mental Health and Well-being in June 2008, as well as the adoption of WHO's Comprehensive Mental Health Action Plan 2013–2020 in May 2013 (European Commission, 2008; House of Representatives, 2008; World Health Organization, 2013b; World Health Organization, 2008). However, between 2014 and 2021, the AAPC for depression and anxiety increased, especially after 2019, potentially linked to mental health issues faced by adolescents during the COVID-19 pandemic due to concerns about virus infection, school closures, and social restrictions (Guessoum et al., 2020; Loades et al., 2020; Santomauro et al., 2021). School closures during the COVID-19 pandemic forced students worldwide to transition to remote learning, with the most negative impacts falling on socioeconomically disadvantaged students (Vogel et al., 2023). The suspension of in-person classes disrupted students' daily routines and limited their access to school resources, elevating stress levels and worsening mental health problems (Lee, 2020). Furthermore, prolonged exposure to computer screens contributed to increased fatigue and mental health symptoms such as headaches, irritability, and anxiety (Nuutinen et al., 2014). During the pandemic, with teachers physically absent, parents played a crucial supportive role in their children's learning process (Guillaume et al., 2022).

In 2021, the incidence of depression in females was 1.6 times that of males, with variations across age groups. Specifically, in the 10–14 age group, females had 1.8 times the incidence of males; in the 15–19 and 20–24 age groups, the ratios were 1.6 and 1.5, respectively. This suggests a decreasing trend in gender disparity with age. Possible explanations include hormonal fluctuations, sociocultural pressures, and changes in mental health awareness. Hormonal changes during puberty may increase the risk of depression among younger females, while increased social support and autonomy with age may help narrow the gender gap (Nolen-Hoeksema, 2012; Faravelli et al., 2013; Keita, 2007; Kuehner, 2017; Oldehinkel and Bouma, 2011; Ridgeway and Correll, 2004). However, over the 32 years, the AAPC for depression and DALYs was higher in males than in females, which may relate to an imbalance in gender ratios and higher societal expectations for males (Gellatly, 2019; Tafuro and Guilamoto, 2020).

The incidence of anxiety among female was 1.5 times higher than that in male in 2021. Compared to males, innate biological factors make females more susceptible to anxiety disorders, with sex hormones—particularly estradiol and progesterone—playing a significant role

in these gender differences (Li and Graham, 2017). Related studies also indicate that anxiety has higher heritability in female than in male (McLean and Anderson, 2009). From a physiological perspective, female brain structure and function differ from males', characterized by relatively larger amygdala volume, heightened sensitivity to threat signals, and distinct neural connectivity patterns in emotion-related brain regions such as the limbic system. These differences result in more intense, prolonged emotional responses to stress and greater predisposition to anxiety (Farhane-Medina et al., 2022). Interestingly, there was no significant gender difference in the AAPC for anxiety over the past 32 years, indicating a similar growth rate of the disease burden in both sexes. Among the three age groups, the AAPC for depression incidence was highest in the 10–14 age group, with females surpassing males. It means the onset of depressive disorders became increasingly younger, which may relate to the significant psychological fluctuations experienced by adolescent females after menarche, leading to an increased risk of major depression (Born et al., 2002). The AAPC for anxiety incidence was highest in the 20–24 age group, with males showing a higher rate than females; however, in 2021, both incidence and DALYs for females aged 20–24 were higher than those for males. This suggests that females face a higher risk of anxiety, while young adult males may experience increasing social pressures as societal transformation and faster-paced living intensify (Akister et al., 2010; Shuman et al., 2022).

The study also found that in 2021, the incidence and DALYs for depression and anxiety were highest in high SDI regions, with the AAPC growth rate over the past 32 years also being the fastest, whereas low SDI regions exhibited the lowest AAPC for incidence and DALYs. This indicates that high SDI regions face greater social and work-related pressures, leading to higher risks for depression and anxiety (Melchior et al., 2007). Conversely, low SDI areas may have lower socioeconomic levels and less focus on mental health, resulting in a higher number of undiagnosed patients (Xiong et al., 2022). Additionally, according to Maslow's hierarchy of needs, individuals in high SDI regions often place greater emphasis on fulfilling psychological needs, particularly those related to social belonging, self-esteem, and self-actualization; when these needs are unmet, the risk of developing depression and anxiety increases (Jorm et al., 2017; Schomerus et al., 2012; Thornicroft, 2007; Xu et al., 2024). This phenomenon aligns with the Easterlin Paradox, which suggests that economic growth does not necessarily lead to increased happiness (Oishi and Kesebir, 2015). It reflects that the changes in socioeconomic conditions and lifestyle have not alleviated the burden of mental health; rather, advancements in technology, increased individualism, and diversification of values have exacerbated the issue (Cai et al., 2019; Saxena et al., 2007). It also indicates that the pressures of modern life are significantly greater than those in traditional societies.

Regionally, North America and Central Latin America exhibited the highest AAPC for depression incidence and DALYs, likely due to high rates of violence, bullying, frequent use of electronic devices, severe income inequality, gender disparities, rapid urbanization, and insufficient understanding of the concept of depressive disorder (Bannink et al., 2014; Errazuriz et al., 2023; Hong et al., 2022). The Andes region of Latin America had the highest AAPC for anxiety incidence and DALYs, similar to the situation in Honduras, where the population may face greater survival pressures, exacerbated by the SARS-CoV pandemic, which intensified economic, social, and familial challenges (Espinoza-Turcios et al., 2023). In contrast, East Asia shows negative growth in both depression incidence and AAPC of DALYs, with the slowest growth in anxiety disorder incidence and DALYs. This may be related to the emphasis on adolescent mental health in these countries over the past 32 years, such as China's "National Mental Health Work Plan (2015–2020)" in 2015 (The State Council of the People's Republic of China, 2015), the "Action Plan for Children's and Adolescents' Mental Health (2019–2022)" in 2019 (The State Council of the People's Republic of China, 2019), and the "Special Action Plan for Strengthening and Improving Mental Health Work for Students in the New Era (2023–

2025)” in 2023 (Ministry of Education of the People's Republic of China, 2023).

From 1990 to 2021, the overall incidence and DALYs for depression and anxiety increased with rising of SDI. High SDI countries exhibited significantly higher incidence and burden compared to low SDI countries, but the AAPC growth rate has declined, possibly due to the better healthcare systems, higher mental health awareness, stronger social support networks, and the implementation of anti-bullying policies in high SDI countries, facilitating early detection, intervention, and treatment, thereby reducing the growth rate of new cases and overall health burden (Dückers and Brewin, 2016; Hong et al., 2022; Lund et al., 2018; Patel and Prince, 2010; Vigo et al., 2016; Yang et al., 2023; Yuchang et al., 2019). Interestingly, localized analysis revealed significant disparities in mental health burdens even within the same SDI group. For instance, some countries in the high SDI group exhibited relatively low incidence and health burdens for these two disorders, while certain low SDI countries showed comparatively high burdens. This suggests that trends can vary dramatically between countries within the same SDI group, potentially related to Simpson's Paradox, where local differences may be obscured when aggregated into an overall trend (Simpson, 2018).

Between 1990 and 2021, the absolute cross-national gap for both diseases increased, indicating that although high SDI countries have advantages in resources, healthcare accessibility, and mental health awareness, their disease burdens have significantly increased. This may relate to higher educational expectations, greater lifestyle changes, increased social pressures, and improved identification of mental health issues (Cifuentes et al., 2008; Oyesanya et al., 2015). Meanwhile, the relative cross-national gap has narrowed, suggesting that the mental health burden may be increasing in low SDI countries. This could be due to cumulative effects of economic, social, and environmental pressures, as well as significant gaps in mental health care services in low SDI countries, where only 12 % of individuals with mental disorders receive the necessary care, compared to 70 % in high-income countries (World Health Organization, 2022). However, this narrowing relative gap may also indicate an imbalanced distribution of global mental health resources and interventions, with high SDI countries experiencing increased burdens in absolute terms but lower growth rates relative to low SDI countries. As far as we know, these findings have not been previously reported. Thus, global health policies need to focus more on the equity of mental health services, especially in resource-scarce regions (Alonge, 2024). Governments should formulate corresponding policies based on actual circumstances to ensure adolescents have access to free and high-quality psychological therapy services. Schools should employ professional psychological counselors to provide regular mental health education and counseling for students. Mental health knowledge training should be conducted for parents to guide them in properly handling conflicts with their children (Jorm et al., 2007). These measures are essential to improve the accessibility and quality of mental health services, thereby addressing the growing global mental health burden.

5. Conclusions

This study found that the incidence and burden of depressive and anxiety disorders significantly increase with rising SDI levels globally. Despite having higher resources and awareness, high SDI countries bear a heavier mental health burden due to high-pressure competition and lifestyle factors. Conversely, the burden in low SDI countries is also gradually increasing with modernization and social change. Furthermore, both local and overall analyses indicate that even within the same SDI group, the trends in disease burden can vary significantly between countries. Compared to 1990, absolute inequality among countries increased by 2021, while relative inequality decreased, indicating that the disparities between countries remain substantial. Therefore, future public health policies should develop more precise and diverse

interventions based on each country's socio-economic conditions and specific population characteristics to effectively reduce the disease burden of depressive and anxiety disorders worldwide.

6. Limitations

The advantages of this secondary analysis include the incorporation of updated data, the use of Joinpoint regression analysis to assess the AAPC of diseases, and the application of the SII and CI to comprehensively evaluate inequality, alongside accurate disease charts for visual representation. However, several limitations should be noted in our study. Firstly, it is constrained by the variability in data quality and missing data from the GBD, particularly from low- and middle-income countries. Secondly, the incidence of depressive and anxiety disorders may be underestimated globally, especially among adolescents and young adults, which could result in bias in the findings. Additionally, this study evaluates depression and anxiety burden based on the SDI, a macro-level measure composed of income, educational attainment, and fertility. However, SDI does not account for individual-level socioeconomic status (SES). Future studies should integrate SES data to explore how socioeconomic status and gender intersect to influence mental health outcomes in adolescents and young adults.

CRedit authorship contribution statement

Zhang Dongjun: Writing – review & editing, Visualization, Supervision, Project administration, Data curation, Conceptualization. **Wu Mingyue:** Writing – original draft, Software, Formal analysis. **Li Xinqi:** Validation, Data curation. **Wang Lina:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wu Jiali:** Writing – review & editing, Visualization, Validation, Software. **Jin Mengyao:** Writing – review & editing.

Ethics approval and consent to participate

Not applicable.

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Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2025.119491>.

Data availability

The data used in this study can be derived from the GBD 2021 (Available at: <https://vizhub.healthdata.org/gbd-results/>).

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